

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457730.6856 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.028 +/- 0.062
Transit depth (Rp/Rs)^2	0.077 +/- 0.34 [%]
Orbital Inclination (inc)	81.22 +/- 2.77
Airmass coefficient 1 (a1)	1.0 +/- 0.19
Airmass coefficient 2 (a2)	-0.018 +/- 0.029
Scatter in the residuals of the lightcurve fit is	18.6 %
Best Comparison Star	#1 - [226, 342]
Optimal Aperture	2.83
Optimal Annulus	6.9
Transit Duration (day)	0.021 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.028 ± 0.062 vs 0.1646 (deviation 1.63σ)
Duration fit vs published	0.021 d vs 0.0758 d (deviation 1.5σ)
Mid-time O–C	-5.6 min
Depth SNR	0.3
Residual scatter	18.6 %

Light curve

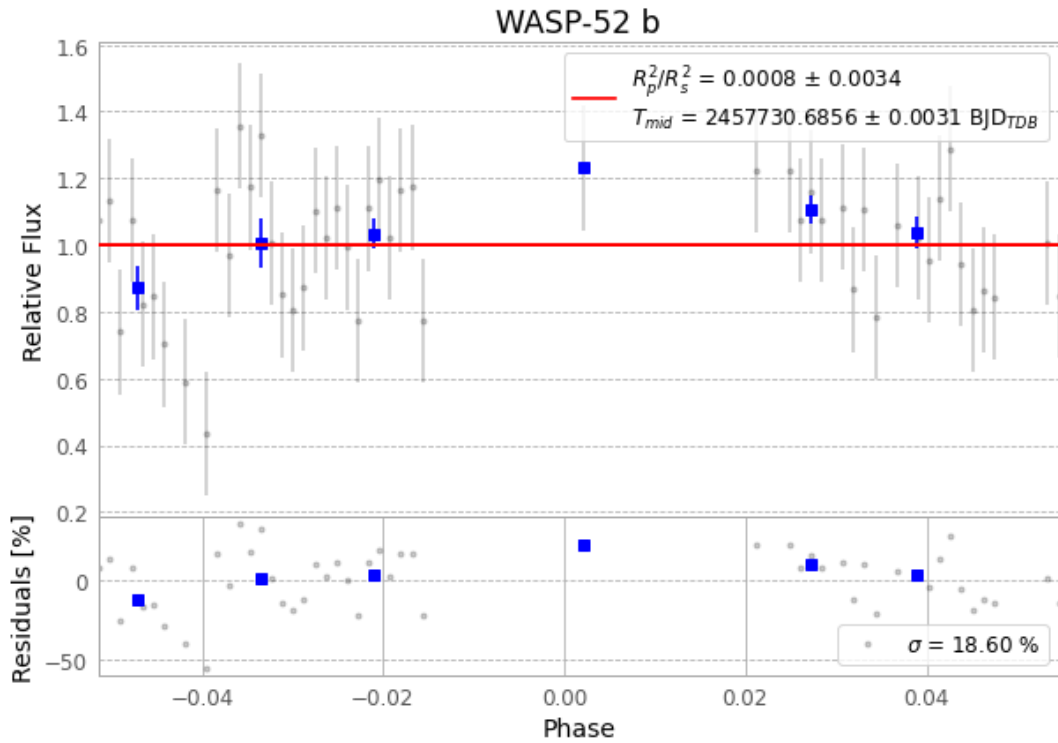


Figure 1 — FinalLightCurve_WASP-52 b_2016-12-07.png (SHA-256 0f3a87d732ac6f82...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 324], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a9d15970e2d0eb24...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

90 FITS frames (59 MB), WASP-52161208021512.FITS → WASP-52161208064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-161208.log (a3b6fd565e5e...), FinalLightCurve_WASP-52 b_2016-12-07.png (0f3a87d732ac...), FinalLightCurve_WASP-52 b_2016-12-07.csv (b58a8c127839...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 02:44Z.